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VPC Peering

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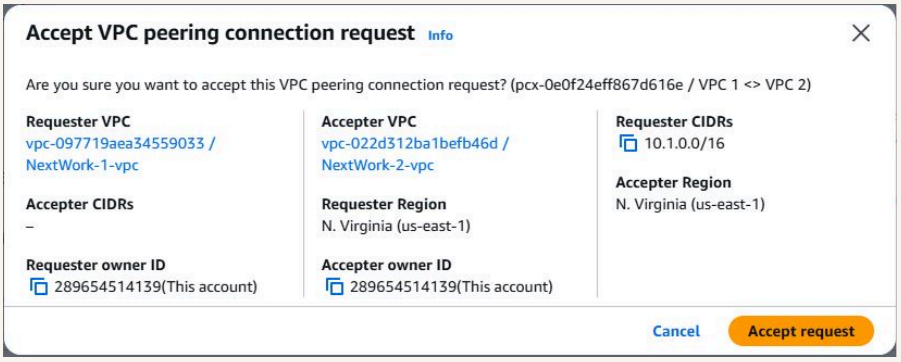


Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private cloud that lets you create and control a secure network within AWS. It is useful because it allows you to isolate resources, manage traffic, and enhance security for your applications.

How I used Amazon VPC in this project

I created two VPCs and connected them using a VPC peering connection so they could communicate privately. I also configured route tables and security settings to enable secure data transfer between them.

One thing I didn't expect in this project was...

I didn’t expect to face connection errors when trying to use EC2 Instance Connect. It helped me understand how important public IPs and correct security group settings are for successful connections.

This project took me...

This project took me 40 minutes to complete.

In the first part of my project...

Step 1 - Set up my VPC

In this step, I will create two VPCs from scratch. I will do that by using the Visual VPC resource map to be able to create the VPC very fast.

Step 2 - Create a Peering Connection

In this step, I will set up a peering connection between the two VPCs I have created. This will allow the two VPCs to connect and communicate with each other.

Step 3 - Update Route Tables

In this step, I will set up the route tables to allow for traffic coming from VPC 1 to VPC 2. And I will set up a way for traffic coming from VPC 2 to VPC 1.

Step 4 - Launch EC2 Instances

In this step, I will launch an EC2 instance in each VPC so I can use them to test the VPC peering connection.

Multi-VPC Architecture

I started my project by launching two VPCs each is one AZ and each has on public subnet. So a total for 2 public subnets across 2 VPCs.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique so the IP addresses of their resources don't overlap. Having overlapping IP addresses could cause routing conflicts and connectivity issues.

I also launched 2 EC2 instances

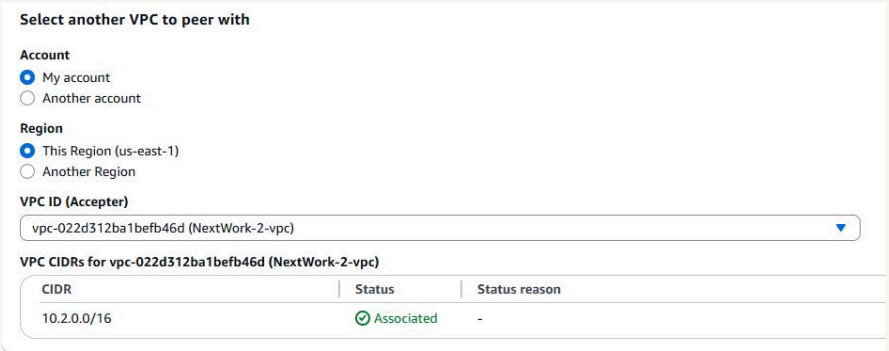
I didn't set up key pairs for these EC2 instances because we can use EC2 instance connect, where AWS stores the key for us, and we can connect to the instance without having to SSH into the instance ourselves.

VPC Peering

A VPC peering connection is a direct connection between two VPCs. It allows the VPCs and their resources to route traffic between them using the private IPs.

VPCs would use peering connections so that data can be transferred between VPCs without going through the public internet.

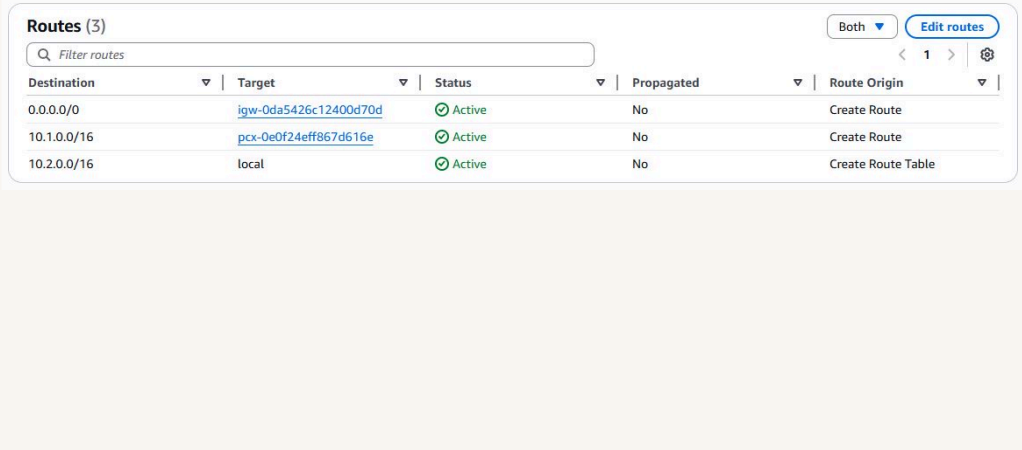
A requester is the VPC that initiates a peering connection; they will send the invitation to connect. The Accepter is the VPC that receives a peering connection request. They can either accept or decline the invitation.



Updating route tables

After accepting a peering connection, my VPCs' route tables need to be updated because VPC 1 needs to know how to get the resources in VPC 2. The route table will direct traffic that will be bound to VPC 2.

My VPCs' new routes have a destination of 10.1.0.0/16 and 10.2.0.0/16. The routes' target was VPC 1 and VPC 2.



In the second part of my project...

Step 5 - Use EC2 Instance Connect

In this step, I will use the EC2 instance to connect to the first EC2 instance and fix any connection issues that arise.

Step 6 - Connect to EC2 Instance 1

In this step, I will use the EC2 instance to connect to the first Instance. And fix any issues or errors that come up while connecting.

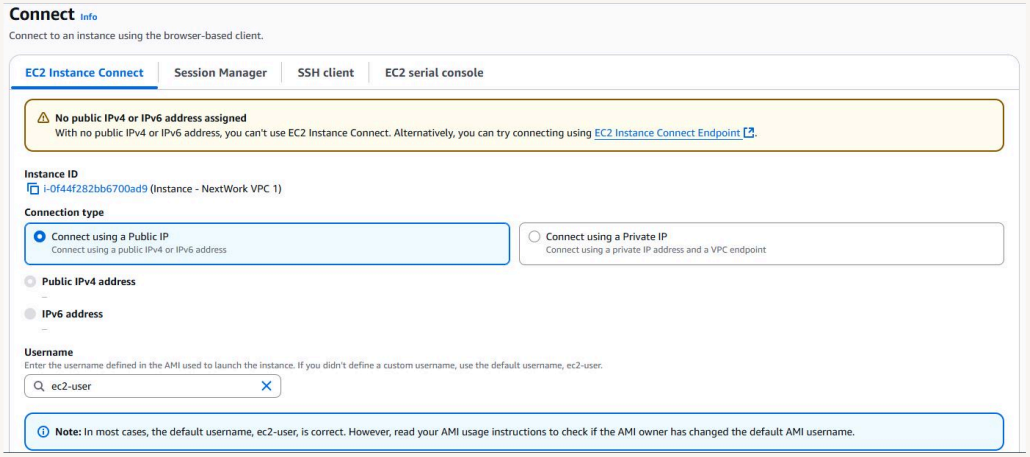
Step 7 - Test VPC Peering

In this step, I will get the first instance to send test messages to the second instance. And then I will fix any issues that come up with the connection between the two instances.

Troubleshooting Instance Connect

Next, I used EC2 Instance Connect to be able to connect to the first EC2 instance.

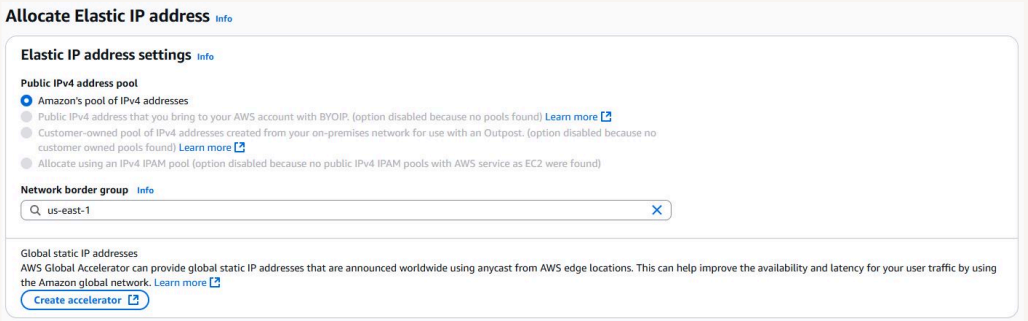
I was stopped from using EC2 Instance Connect because there was no public IP for the instance, because we had the setting disabled for automatically assigning public IPs.



Elastic IP addresses

To resolve this error, I set up Elastic IP addresses. Elastic IP addresses are static IPv4 addresses that get allocated to your AWS account.

Associating an Elastic IP address resolved the error because it gave the instance a public IP that we can connect to through the EC2 instance connect.

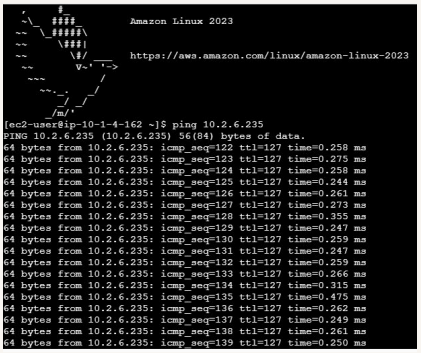


Troubleshooting ping issues

To test VPC peering, I ran the command ping. Ping is a common computer network tool used to check whether your computer can communicate with another computer or device on a network.

A successful ping test would validate my VPC peering connection because it shows the two instances are connecting and are able to communicate with each other.

I had to update my second EC2 instance's security group because the ping command was not working. I added a new rule that allowed ICMP traffic from VPC 1's CIDR block.



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